Ruomeng Ding

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Research interests: LLM Agents & Reasoning · Preference Learning & Elicitation · Reliable LLM Evaluation · AI4AI

EDUCATION

Ph.D. in Computer Science, University of North Carolina at Chapel Hill — Aug 2025 – Present

- Advised by Prof. Zhun Deng.

M.S. in Computer Science, Georgia Institute of Technology — Aug 2022 – May 2025

- GPA: 4.00/4.00; specialization: Machine Learning.

B.Eng. in Computer Science and Technology, Tianjin University — Aug 2018 – May 2022

- GPA: 3.75/4.00; major rank: 11/169 (top 6.5%).

PUBLICATIONS

** Equal contribution. Workshop papers are identified explicitly.*

LLM Agents & Reasoning

1. [Preprint] [AutoMLR Workshop @NeurIPS 2026] **R. Ding**, Q. Shen, Z. Han, M. Chen, D. Chen, and Y. Li, "Agentic Data Engineering for LLMs: System Design and a Controlled Empirical Study," in *Workshop for Autonomous Machine Learning Research at NeurIPS*, 2026. arXiv:2610.14238 [paper] [code]
2. [AAAI 2026] **R. Ding**, W. Cheng, M. Shao, and C. Zhao, "SkillGen: Learning Domain Skills for In-Context Sequential Decision Making," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026 **(Oral)** [paper] [code]
3. [ACL 2024] **R. Ding**, C. Zhang, L. Wang, Y. Xu, M. Ma, W. Zhang, S. Qin, S. Rajmohan, Q. Lin, and D. Zhang, "Everything of Thoughts: Defying the Law of Penrose Triangle for Thought Generation," in *Findings of the Association for Computational Linguistics (ACL)*, 2024 [paper] [code]

Preference Learning & Elicitation

4. [Preprint] [FLLMPT Workshop @NeurIPS 2026] **R. Ding***, T. Gao*, L. Geng, M. Bansal, and Z. Deng, "Who Should We Listen to More? Welfare-Aware Preference Acquisition for Pluralistic Alignment," in *Workshop on Foundations of LLM Post-Training in Changing Environments at NeurIPS*, 2026. arXiv:2610.19035 [paper] [code]
5. [ICML 2026] **R. Ding***, T. Gao*, T. P. Zollo, E. Bachmat, R. Zemel, and Z. Deng, "Whom to Query for What: Adaptive Group Elicitation via Multi-Turn LLM Interactions," in *The 43rd International Conference on Machine Learning (ICML 2026)*, 2026 [paper] [code]

Reliable LLM Evaluation

6. [Preprint] [TAE Workshop @NeurIPS 2026] Q. Shen, X. Li, **R. Ding**, Y. Chen, D. Chen, and Y. Li, "Speculative Evaluation of Stochastic LLMs," in *Can We Trust AI Evaluation? Workshop at NeurIPS*, 2026. arXiv:2609.28560 [paper] [code]
7. [COLM 2026] **R. Ding***, Y. Pang*, H. Sun, Y. Wang, Z. S. Wu, and Z. Deng, "Rubrics as an Attack Surface: Stealthy Preference Drift in LLM Judges," in *Proceedings of the 3rd Conference on Language Modeling (COLM)*, 2026 [paper] [code]
8. [NeurIPS 2024] R. Yang, **R. Ding**, Y. Lin, H. Zhang, and T. Zhang, "Regularizing Hidden States Enables Learning Generalizable Reward Model for LLMs," in *The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024 [paper] [code]

Reliable Machine Learning & Applications

9. [SDM 2025] **R. Ding**, X. Zhao, C. Zhao, M. Shao, Z. Chen, and H. Chen, "Evidence-Based Out-of-Distribution Detection on Multi-Label Graphs," in *Proceedings of the 2025 SIAM International Conference on Data Mining (SDM)*, pp. 588–597, SIAM, 2025 [paper]
10. [FSE 2023] **R. Ding**, C. Zhang, L. Wang, Y. Xu, M. Ma, X. Wu, M. Zhang, Q. Chen, X. Gao, X. Gao, H. Fan, S. Rajmohan, Q. Lin, and D. Zhang, "TraceDiag: Adaptive, Interpretable, and Efficient Root Cause Analysis on Large-Scale Microservice Systems," in *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2023 [paper] [slides]
11. [KDD 2023] L. Wang, C. Zhang, **R. Ding**, Y. Xu, Q. Chen, W. Zou, Q. Chen, M. Zhang, X. Gao, H. Fan, S. Rajmohan, Q. Lin, and D. Zhang, "Root Cause Analysis for Microservice Systems via Hierarchical Reinforcement Learning from Human Feedback," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2023 [paper]
12. [VLDB 2023] Y. Chen, C. Zhang, M. Ma, Y. Liu, **R. Ding**, B. Li, S. He, S. Rajmohan, Q. Lin, and D. Zhang, "ImDiffusion: Imputed Diffusion Models for Multivariate Time Series Anomaly Detection," in *Proceedings of the VLDB Endowment (VLDB)*, 2023 [paper] [code]

RESEARCH EXPERIENCE

Research Intern, **Alibaba Token Foundry (Tongyi Lab)** May 2026 – Aug 2026
Hangzhou, China (Onsite)

Advised by Dr. Qianli Shen and Daoyuan Chen.

- Investigated agent-driven data selection, reward design, and curriculum learning for LLM training.
- Developed Agentic Data Engineering to refine strategies using cumulative experimental feedback; evaluated generalization across four held-out benchmarks.

Research Intern, **Microsoft Research** May 2024 – Aug 2024
Redmond, WA (Onsite)

Advised by Dr. Minghua Ma and Dr. Ze Li.

- Developed LLM-based multi-agent systems for incident triage.
- Designed methods, conducted experiments, and developed prototypes.

Research Assistant, **University of Illinois Urbana-Champaign** Mar 2024 – Jul 2024
Urbana, IL (Remote)

Advised by Dr. Tong Zhang, working with Rui Yang.

- Studied reward model generalization under distribution shifts for reinforcement learning from human feedback (RLHF); second author of a NeurIPS 2024 paper.
- Conducted literature review and RLHF experiments; contributed to paper writing and the code repository.

Research Intern, **Microsoft Research Asia** Nov 2022 – Aug 2023
Beijing, China (Hybrid)

Advised by Dr. Lu Wang and Dr. Chaoyun Zhang.

- Improved LLM reasoning with domain knowledge and Monte Carlo Tree Search; led methodology, experiments, writing, and code release (Findings of ACL 2024, first author).
- Developed RL- and RLHF-based root cause analysis for large-scale microservice systems, with deployment in M365 (KDD 2023, third author; ESEC/FSE 2023, first author).

Research Assistant, **Tianjin University** Aug 2021 – Aug 2022
Tianjin, China (Onsite)

Advised by Dr. Minglai Shao and Dr. Wenjun Wang.

- Studied multi-label out-of-distribution detection on graphs, co-advised by Dr. Xujiang Zhao and Dr. Chen Zhao (UDM Workshop at AAAI 2023, first author).
- Conducted experiments on dynamic community detection on graphs (IEEE Transactions on Cybernetics, 2023).

HONORS AND FELLOWSHIPS

Doctoral Merit Fellowship, University of North Carolina at Chapel Hill 2025–2026

Merit Scholarship, Georgia Institute of Technology 2022–2023

ACADEMIC SERVICE

- Conference reviewer: AAAI 2026–2027, ICLR 2026–2027, COLM 2026, NeurIPS 2026, ICML 2026, KDD 2025–2026.